

mechanically verifiable -> reusable as an RL reward

model-graded -> who grades the grader?

exact match

string / choice match
≈ free per item

answer key is the grader

programmatic

tests · checkers · env state
build a harness once

ABC: 7/10 graders unsound

LLM judge

holistic model judge
one model call

GPT-4<->human 85% (humans 81%)

rubric + judge

per-item rubric + judge
experts author criteria

HealthBench: 48,562 criteria

human

expert / preference grading
expert-hours per item

GDPval: >1 h per grade

cost to grade one item -> (proxy metrics — BLEU/ROUGE, n=36 — sit outside the ladder: cheap but no longer trusted)